

Quiz 1 Solution

Ans 1:

An agent is anything that can perceive its environment and act on it.

Core Role: Bridge between environment and intelligent behaviour.

Environment → [Agent box] → *Environment*

The agent gets input (percepts) from its environment and takes actions that affect it.

Cycle:

Perception → Decision → Action → Environment

Example:

A cleaning robot sees dirt (perception), moves to it (action).

Ans 2:

To solve this using **Breadth-First Search (BFS)**, we explore the graph layer by layer, starting from node **A** (the root) and moving outward to its neighbors. BFS uses a **Queue (FIFO)** data structure.

Step	Current Node	Queue (Front → Rear)	Visited List	Action
1	-	[A]	{}	Initial state: Start at A .
2	A	[B, C]	{A}	Dequeue A . Enqueue neighbors B and C .
3	B	[C, D, E]	{A, B}	Dequeue B . Enqueue neighbors D and E .
4	C	[D, E, F, G]	{A, B, C}	Dequeue C . Enqueue neighbors F and G .

Step	Current Node	Queue (Front → Rear)	Visited List	Action
5	D	[E, F, G, H]	{A, B, C, D}	Dequeue D . Enqueue neighbor H .
6	E	[F, G, H]	{A, B, C, D, E}	Dequeue E . (H already in visited).
7	F	[G, H]	{A, B, C, D, E, F}	Dequeue F . (H already in visited).
8	G	[H]	{A, B, C, D, E, F, G}	Dequeue G . (H already in visited).
9	H	[]	{A, B, C, D, E, F, G, H}	Dequeue H . Goal Reached!

Path:

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow H$

Final Path:

$A \rightarrow B \rightarrow D \rightarrow E \rightarrow H$

(Note: $A \rightarrow B \rightarrow E \rightarrow H$, $A \rightarrow C \rightarrow F \rightarrow H$, and $A \rightarrow C \rightarrow G \rightarrow H$ are also valid paths of the same shortest length, but the standard left-to-right BFS discovery order typically picks the first one encountered.)

Time Complexity:

$O(b^d) = O(2^3) = 8$